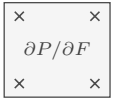
(a) formats of the tangent

more of the tangent computed once per Newton step, less in each application









	matrix-free	**quadrature-point store**	**element matrix**	**assembled matrix**
set-up, per Newton step	nothing stored	pulled-back tangent $\widehat{S}^q$ at the points, encoded	$\widehat{S}^q$ contracted with the basis into K_e, encoded	K_e summed into the node pattern, encoded
application, per Krylov iteration	gradients, material law and its derivative at every point	gradients at the points, decode, 9×9 contraction	decode, dense product per element	decode, one block product per row

(b) measures on the stored formats

exact: the application is unchanged

approximate

